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TITLE OF THE EXPERIMENT

To create a JavaScript-based to-do list application using DOM manipulation, dynamic HTML element creation, and event handling.

DESCRIPTION

This program creates an interactive To-Do List application using HTML, CSS, and JavaScript. The interface provides a text input field and an "Add Task" button that lets the user enter and submit new tasks, either by clicking the button or pressing the Enter key. Each task is dynamically created as a list item using DOM methods such as `createElement()` and `appendChild()`, and every task item includes a "Complete" button that toggles a strikethrough style on the task text, and a "Delete" button that removes the task from the list. An empty-state message ("No tasks available.") is shown or hidden automatically based on whether the task list is empty. Basic input validation ensures an alert is shown if the user tries to add a blank task. CSS styling is applied to give the application a clean, card-based layout with clear visual feedback for completed and active tasks. Thus, the program demonstrates DOM selection, dynamic HTML element creation, event handling, and interactive UI updates.

PROGRAM

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>My To-Do List</title>
```

<style>

/* Page styling */

```
body{  
    font-family: Arial, sans-serif;  
    background-color: #f2f2f2;  
    margin: 0;  
    padding: 40px;  
}
```

/* Main container */

```
.container {  
    width: 500px;  
    margin: auto;  
    background-color: white;  
    padding: 25px;  
    border-radius: 10px;  
    box-shadow: 0 0 10px gray;  
}
```

/* Heading */

```
h1 {  
    text-align: center;  
}
```

/* Input box */

```
#taskInput {
```

```
width: 65%;  
padding: 10px;  
border: 1px solid #ccc;  
border-radius: 5px;  
font-size: 16px;  
}
```

```
/* Add button */  
#addBtn {  
    padding: 10px 15px;  
    background-color: green;  
    color: white;  
    border: none;  
    border-radius: 5px;  
    cursor: pointer;  
}
```

```
/* Task list */  
#taskList {  
    margin-top: 20px;  
    padding: 0;  
}
```

```
/* Individual task */  
.task {  
    display: flex;  
    align-items: center;  
    justify-content: space-between;
```

```
padding: 12px;
margin-bottom: 10px;

background-color: #f8f8f8;

border: 1px solid #ddd;
border-radius: 5px;
}

/* Task text */
.task-text {
    flex: 1;
    font-size: 16px;
}

/* Completed task */
.completed {
    text-decoration: line-through;
    color: gray;
}

/* Buttons inside task */
.complete-btn,
.delete-btn {
    padding: 7px 10px;
    margin-left: 5px;
    border: none;
```

```
    border-radius: 4px;
    cursor: pointer;
}
```

```
.complete-btn {
    background-color: blue;
    color: white;
}
```

```
.delete-btn {
    background-color: red;
    color: white;
}
```

```
/* Empty message */
#emptyMessage {
    text-align: center;
    color: gray;
    margin-top: 20px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h1>My To-Do List</h1>
```

```
<!-- Input and Add button -->
```

```
<input  
  type="text"  
  id="taskInput"  
  placeholder="Enter a task....."  
>
```

```
<button id="addBtn">  
  Add Task  
</button>
```

```
<!-- Empty message -->
```

```
<p id="emptyMessage">  
  No tasks available.  
</p>
```

```
<!-- Task list -->
```

```
<div id="taskList"></div>
```

```
</div>
```

```
<script>
```

```
// Select the required HTML elements

const taskInput = document.getElementById("taskInput");

const addBtn = document.getElementById("addBtn");

const taskList = document.getElementById("taskList");

const emptyMessage =
    document.getElementById("emptyMessage");

// Function to update the empty message

function updateEmptyMessage() {

    if (taskList.children.length === 0) {

        emptyMessage.style.display = "block";

    } else {

        emptyMessage.style.display = "none";

    }

}
```

```
// Add Task button click event

addBtn.addEventListener("click", function() {

    // Get the task entered by the user

    const taskText = taskInput.value.trim();

    // Check if the input is empty

    if (taskText === "") {

        alert("Please enter a task.");

        return;

    }

    // Create the main task container

    const task = document.createElement("div");

    task.className = "task";

    // Create the task text

    const taskTextElement =

        document.createElement("span");
```



```
taskTextElement.className = "task-text";
```

```
taskTextElement.textContent = taskText;
```

```
// Create Complete button
```

```
const completeBtn =
```

```
    document.createElement("button");
```

```
completeBtn.className = "complete-btn";
```

```
completeBtn.textContent = "Complete";
```

```
// Create Delete button
```

```
const deleteBtn =
```

```
    document.createElement("button");
```

```
deleteBtn.className = "delete-btn";
```

```
deleteBtn.textContent = "Delete";
```

```
// Complete button event
```

```
completeBtn.addEventListener("click", function() {
```

```
    taskTextElement.classList.toggle("completed");
```

```
});
```

```
// Delete button event
```

```
deleteBtn.addEventListener("click", function() {
```

```
    task.remove();
```

```
    updateEmptyMessage();
```

```
});
```

```
// Add elements to task container
```

```
task.appendChild(taskTextElement);
```

```
task.appendChild(completeBtn);
```

```
task.appendChild(deleteBtn);
```

```
// Add task to task list
```

```
taskList.appendChild(task);
```

```
// Clear input box
```

```
taskInput.value = "";
```

```
// Update empty message

updateEmptyMessage();

});

// Display empty message initially

updateEmptyMessage();

</script>

</body>

</html>
```

OUTPUT

